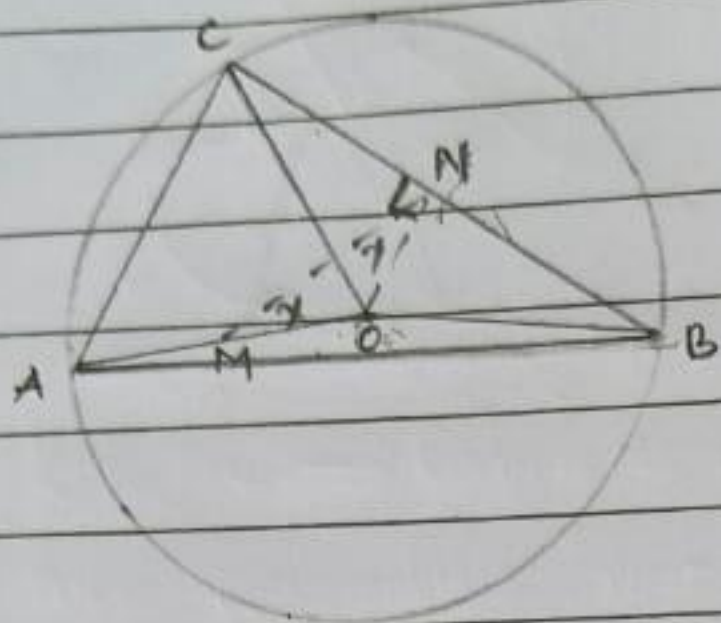




$$\angle OMN = x$$

$$\angle ABC = 4\angle OMN$$

$$= 4x$$

$$\angle ACB = 6\angle OMN$$

$$= 6x$$

$$x = \dots ?$$

Jawab:

- $\angle ONC = 90^\circ$ (karena membagi garis CB menjadi sama panjang / membaginya di tengah)

- $\angle CON = \frac{1}{2} \angle BOC$

- $\angle BAC + \angle ABC + \angle ACB = 180^\circ$

$$\angle BAC + 4x + 6x = 180^\circ$$

$$\angle BAC = 180^\circ - 10x$$

- $\angle AOC = 2\angle ABC$

$$= 2 \cdot 4x$$

$$= 8x$$

} hub. sudut pusat dan keliling

- $\angle AON = \angle AOC + \angle CON$

$$= 8x + \frac{1}{2} \angle BAC$$

$$= 8x + (180^\circ - 10x)$$

$$= 180^\circ - 2x$$

} $\Rightarrow$ Hub sudut pusat dan keliling kemudian dibagi 2 karena ada garis pembagi

$$\angle AON = \angle MON = 180^\circ - 2x$$

$$\angle OMN = \angle ONM = x \text{ (segitiga sama kaki)}$$

$$OC = OA = OB = \text{jari} - \text{jari} = \text{panjangnya sama, sehingga}$$

$$OM = ON = \frac{1}{2} OC = \frac{1}{2} r$$

$$\Rightarrow \cos \angle CON = \frac{\text{samping}}{\text{miring}}$$

$$= \frac{OM}{OC}$$

$$= \frac{\frac{1}{2} r}{r}$$

$$= \frac{1}{2} \Rightarrow 60^\circ$$

$$\text{Jadi, } \angle CON = 180^\circ - 10x$$

$$60^\circ = 180^\circ - 10x$$

$$60^\circ + 10x = 180^\circ$$

$$10x = 180^\circ - 60^\circ$$

$$10x = 120^\circ$$

$$x = 12$$